

Duration judgments with conflicting audiovisual cues

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How does the brain integrate conflicting temporal information across sensory modalities? Previous work has shown that observers integrate audiovisual duration cues optimally when the conflict between the two cues is small. Does causal inference lead to a breakdown of audiovisual integration when duration conflicts are large? Here, we address this question by testing a wide range of cue conflicts in an auditory duration-discrimination task. In the bimodal task, participants compared the auditory durations of a test and a standard stimulus. Audiovisual durations were consistent in the test stimulus, but differed by seven conflict durations (± 250 ms) in the standard stimulus. Auditory durations were tested at two levels of noise. We found that auditory duration percepts shifted systematically toward the visual duration, especially with high auditory noise. The shift was proportional to the cue-conflict magnitude, inconsistent with the predictions of causal inference. We compared several models. A heuristic model in which the observer probabilistically switches between the visual and auditory cues was preferred for most participants, although performance differences across models were small. Model simulations further revealed that given the measured sensory noise, competing models can be discriminated only with unreasonably large conflicts. Within the conflict range tested here, all models produced largely overlapping, near-linear shifts as a function of the cue conflict magnitude. Our results suggest that observers do not rely on causal inference when judging auditory durations under our conditions. Although the heuristic cue-switching model was favored by model comparison, high sensory variability in auditory duration encoding—particularly under high auditory-noise conditions—limits the discriminability of competing computational models. Under such levels of encoding noise, forced fusion, causal inference, and probabilistic switching models generate highly similar predictions across the experimentally tested conflict range.

Keywords: audiovisual integration; duration perception; causal inference; cue conflict; multisensory perception; psychophysics

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Introduction

Accurately estimating the timing of events is essential for many everyday behaviors, from playing a musical instrument to

catching a fly ball. Humans rely on multiple sensory modalities to perceive the timing and duration of events in the world.

Prevailing models of timing mechanisms often rely on a clock analogy, particularly pacemaker–accumulator models (Treisman, 1963; Gibbon, 1977; Wittmann, 2009; Grondin, 2014; Hartcher-O’Brien et al., 2016; Grondin, 2025). These models propose that the brain contains an internal clock, where a pacemaker emits regular pulses that are accumulated over time to

estimate duration. However, there is no concrete evidence for the existence of such a mechanism in the brain. While these models successfully explain timing properties such as Vierordt's law (Lejeune and Wearden, 2009; Jazayeri and Shadlen, 2010; Mamassian and Landy, 2010), which accounts for overestimation of short durations and underestimation of long durations, neural response patterns predicted by these models do not align well with actual neural data (Buonomano and Laje, 2010; Muller and Nobre, 2014; Kononowicz and van Wassenhove, 2016). Unlike vision or audition, however, time does not have a dedicated sensory receptor; rather, temporal perception is inferred based on signals elicited from multiple modalities.

Despite the inherent difficulty of temporal estimation, the brain must operate with remarkable precision and consistency when timing our actions, which is crucial for effective behavior. Consider ensemble music performance: after hearing a musical phrase, you must time your entrance so it coincides with the other performers, and entering too early or too late can disrupt the coordination of the group. Achieving this kind of temporal precision relies not only on motor coordination but also on the brain's ability to accurately perceive and integrate temporal cues from multiple senses.

In natural environments, sounds and sights rarely occur in isolation; temporal judgments should often be made across modalities. Continuing our musical example, estimation of when you should begin to play can come both from hearing the violinist and from seeing the movements of her bow. This multisensory encoding makes time perception inherently more variable and harder to investigate. The brain estimates time using cues embedded in visual, auditory, and possibly from other sensory inputs, each with its own level of precision and bias. Even when stimuli are relatively clear (low external noise), differences in the physical properties and neural conduction velocities of vision and audition create inherent temporal discrepancies. Light travels faster than sound, but paradoxically, auditory signals often reach the cortex more quickly than visual ones due to faster transduction and shorter conduction delays (Zoefel and VanRullen, 2017). Auditory stimuli typically arrive at central processing regions within 8–10 ms, whereas visual signals take approximately 20–40 ms to reach the cortex from the retina (King and Palmer, 1985; Vroomen and Keetels, 2010). In this study, we investigated how the brain perceives durations when presented with conflicting

information across modalities, specifically examining whether and how observers integrate auditory and visual timing cues under such conflict.

When sensory information is available from multiple modalities, the brain faces the challenge of integrating these cues into a single percept. Research across diverse sensory systems has shown that this integration is not arbitrary. Rather, the brain tends to weight each cue according to its reliability. This strategy—known as statistically optimal cue integration—helps minimize the total uncertainty of the fused percept (Maloney and Landy, 1989; Young et al., 1993; Landy et al., 1995; Ernst and Banks, 2002). This optimal cue-integration strategy has been documented across a wide array of sensory tasks, including vision-touch integration for object slant estimation (Ernst and Banks, 2002; Hillis et al., 2002), taste-smell combination for flavor perception (Maier and Elliott, 2020), and audiovisual integration for spatial localization (Alais and Burr, 2004; Negen et al., 2018). These findings underscore the generality of Bayesian cue combination as a core principle of sensory processing (Landy and Kojima, 2001; Landy et al., 2011; Wolpert et al., 2011; Alais and Burr, 2019; Rossi et al., 2021), providing a robust normative framework for understanding multisensory perception.

In previous experiments (Shams et al., 2005; Hartcher-O'Brien et al., 2014), sensory cues across modalities were deliberately aligned in space and time, or only slightly mismatched, effectively biasing the brain toward assuming a single source. The use of low-conflict conditions is a way of examining how different cues are combined with respect to their reliabilities. Under such low-conflict conditions, it is reasonable to assume a common cause and therefore integrate the cues. However, everyday perception often presents greater cue disparities, whether from environmental factors (e.g., room acoustics), technological delays (e.g., audio–video lag), or neural processing differences. In those cases, the brain must address two interrelated questions: Are these disparate cues from the same source? And if so, how should they be combined? Rather than integrating across the board, the brain may dynamically shift from fusion to segregation depending on the degree of mismatch, an insight that has prompted the development of Bayesian causal-inference models. These models explain how integration diminishes as spatial or temporal disparity grows, as notably demonstrated in

the ventriloquist-illusion paradigm (Shams et al., 2005; Körding et al., 2007; Hong et al., 2021). Under this account, integration is conditional on inferred causal structure rather than obligatorily applied to all instances of cue combination (Körding et al., 2007; Badde et al., 2018, 2021; Hong et al., 2021; Badde et al., 2023; Li et al., 2025).

While causal inference has been implicated in multisensory perception across various sensorimotor tasks, much less is known about whether it operates in temporal perception, particularly in duration judgments. Bayesian causal-inference models posit that perception emerges from a combination of common-cause inference and likelihood-based integration; that is, the brain infers whether different sensory cues share a source and, if so, integrates them accordingly. Hartcher-O'Brien et al. (2014) provided evidence for optimal cue integration in duration perception under low-conflict conditions: when audiovisual signals specified the same duration, observers combined duration information in a statistically optimal fashion. However, it remains unexplored whether visual and auditory duration are still integrated when the conflict between them becomes substantial, as Bayesian causal inference predicts a breakdown of cue integration under large-conflict conditions.

In the present study, we investigated how observers judge auditory duration when presented with conflicting audiovisual (AV) cues. We manipulated two key factors: the degree of cross-modal cue conflict and auditory cue reliability. Before introducing audiovisual cue conflict, we measured each participant's audiovisual temporal bias in a cross-modal duration-discrimination task and compensated for it in the bimodal experiment. Successful manipulation of auditory cue reliability was confirmed in a separate unimodal experiment. Although participants were instructed to base their judgments on the auditory signal alone, perceived auditory duration was systematically biased toward the task-irrelevant visual duration, especially when the auditory cue was less reliable. Moreover, we observed that visual-induced biases in auditory duration judgments increased approximately linearly with audiovisual conflict magnitude. This pattern differs from the non-linear breakdown of integration predicted by traditional Bayesian causal-inference models. Instead, the observed behavior was consistent with a model in which observers flexibly switched between reliance on auditory and visual cues across trials. Model comparison further suggested that, under substantial sensory uncertainty,

multiple computational strategies produce highly similar behavioral predictions, limiting the discriminability of competing integration models.

With well-controlled experiments and a systematic evaluation of computational models, this work characterizes auditory duration perception under substantial audiovisual cue conflict and highlights the challenges in studying multisensory time perception.

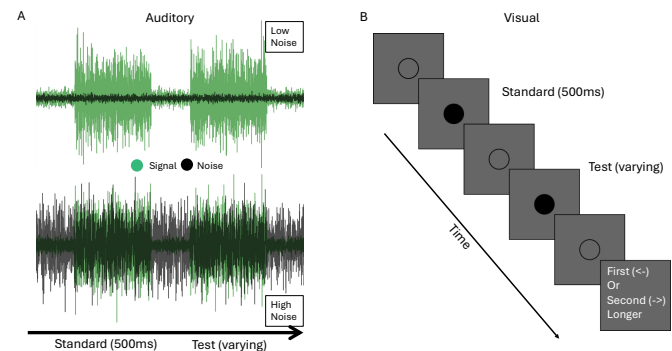


Figure 1: Experimental stimuli and trial procedure. Schematic illustration of the auditory and visual stimuli used in the duration-discrimination tasks. **A** Example trials of the unimodal auditory task under the low (top) and high (bottom) auditory-noise conditions. In the low-noise condition, the auditory signal (bright green) is clearly distinguishable from the background noise (black). In the high-noise condition, signal and noise overlap substantially. **B** Trial timeline for the unimodal visual task. A disk outline remained visible throughout the trial to provide a stable visual reference. Each trial consisted of two sequentially presented stimulus durations (a standard and a test), separated by a short inter-stimulus interval (ISI). During each signal duration, the disk was filled (signal-on; shown dark), indicating the presence of the visual stimulus. Pre- and post-stimulus periods flanked the two intervals. The 500-ms standard and an adaptively varied test duration were presented in random order (standard-first order shown). Participants reported whether the first or second duration was longer.

Methods

Participants

We recruited twelve participants for the experiment, two of whom were the authors O.Y. and L.N., two were other lab members, while the rest were naive to the purposes of the experiment but experienced in psychophysical experiments. Testing was conducted in accordance with the ethical standards approved by the Institutional Review Board at New York University (pro-

tocol number FY2016–595). All participants provided informed consent prior to their involvement in the study and were compensated \$15 per hour for participation, except for the first author.

Apparatus

All experiments were conducted on a 21-inch CRT (Trinitron GDM-5402, Sony, Japan) with a refresh rate of 120 Hz and a spatial resolution of 800x600 pixels. The experiments were designed and executed using the Python PsychoPy package (Peirce, 2007). The participants sat in a darkened room, approximately 60 cm away from the computer screen, with their head stabilized by a chin rest.

Stimuli

Visual stimuli

The visual stimulus was a black disk (0.07 cd/m^2) displayed in the center of the screen on a gray background (8.8 cd/m^2). The width of the disk was 3 degrees of visual angle. Throughout each trial the disk's black outline was continuously displayed. During the stimulus intervals, the inside of the circle changed from the background gray to black.

Auditory stimuli

We adapted the stimuli from Hartcher and colleagues (2014) and Shi and colleagues (2013) in which they created their stimuli by adding a pure-tone signal to a pink-noise background and manipulated duration uncertainty by varying the signal-to-noise ratio. Compared to their manipulation, in our stimulus, we generated white-noise waveforms for both the background and the signals. The background signal had a constant amplitude throughout the trial (Figure 1). The waveform for the signals had a low, non-zero amplitude before, between and after the pair of standard and test stimuli. During the standard and test, the amplitude was increased. We then differentiated the signal from the background using different broadband filters to ensure that the signal was always detectable even when the background noise was louder than the stimuli.

The auditory signals and the background noise were both filtered white-noise signals sampled at 48 kHz. Each interval's stimulus was a noise burst with increased amplitude modulated by a raised cosine (5 ms cosine ramp up, flat peak, cosine ramp down). The signal was embedded in a continuous background noise that

spanned the entire trial. The signals waveform consisted of a pre-stimulus interval, first-interval stimulus, inter-stimulus interval, second-interval stimulus and post-stimulus interval, with the stimulus intervals defined by the raised-cosine amplitude envelope.

We manipulated auditory uncertainty by varying the relative amplitude of the continuous background noise with respect to the signal waveform before finally summing the two waveforms. In the low-noise condition, the background amplitude was approximately 0.1 times the stimulus amplitude, whereas in the high-noise condition it was approximately 1.2 times that amplitude. The final mixed waveform was then peak-normalized for playback.

To keep the “signal” perceptually distinct even with the high noise level, we applied different band-pass filters to the signal and background tracks. The background waveform was band-pass filtered with cutoffs at 10 and 610 Hz. The signal waveform (including the test and standard) was band-pass filtered from 150–775 Hz to make the stimuli more distinguishable from the background. The two waveforms were then summed to create the final stimulus. All filtering was implemented using second-order sections (SOS) Butterworth filters with a filter order of 4.

Auditory stimuli were calibrated using a Brüel & Kjær Type 2240 sound level meter positioned at the participant's head location. Sound levels were measured as A-weighted equivalent continuous levels (LAeq). The test and standard stimuli had an average intensity of approximately 70.5 dB SPL (from the sum of the stimuli and background). In the low-noise condition, the continuous background noise (the sum of the background waveform and the non-stimulus signals waveform) was presented at 57.4 dB SPL, yielding a signal-to-noise ratio (SNR) of approximately +13.1 dB. In the high-noise condition, the background noise was presented at 67.5 dB SPL, yielding an SNR of approximately +2.9 dB. All sound intensities are reported as A-weighted levels.

To further characterize the stimulus structure, we additionally measured the sound levels of the individual stimulus components prior to mixing the background and signals waveforms. The background waveform averaged approximately 51.5 dB SPL in the low-noise condition and 67.0 dB SPL in the high-noise condition. For the signals waveform, the level during non-stimulus intervals (pre-stimulus, ISI, post-stimulus)

was approximately 57.1 dB SPL in the low-noise condition and 54.5 dB SPL in the high-noise condition. During the stimulus intervals, levels increased to approximately 70.6 dB SPL in the low-noise condition and 66.6 dB SPL in the high-noise condition. These measurements confirm that the stimulus contained a clearly detectable signal embedded in background noise, with the signal interval producing a distinct increase in sound level relative to the baseline.

Procedure

Participants completed four experiments. First, they completed unimodal auditory and unimodal visual tasks, which were used to measure the uncertainty of duration perception for each modality. They carried out these two tasks in separate blocks at the beginning of the experiment to become familiar with the stimuli.

After completing the unimodal tasks, participants completed the cross-modal baseline task, allowing us to measure and correct for perceptual bias between auditory and visual stimulus durations. We excluded one participant who showed a substantial perceptual bias (560 ms) toward visual duration in the high-noise auditory condition.

Finally, participants performed the main experiment, which consisted of three sessions of approximately one hour each. Each session started with a flexible number of practice trials in which participants received feedback about the correctness of their answers until they felt comfortable with each task.

All tasks used a 2IFC design ("Which interval was longer?"). In each trial there was a standard and test stimulus. The order of standard/test stimuli was randomized across trials. Each trial comprised a pre-stimulus noise, first stimulus, an inter-stimulus interval (ISI), second stimulus, and a post-stimulus noise. Short breaks occurred every ~30 trials.

In the unimodal auditory, unimodal visual, and cross-modal experiments, the standard duration was fixed at 500 ms. In the bimodal experiment, the auditory component of the standard interval was always 500 ms. When audiovisual conflict was 0 ms, the visual component was also nominally 500 ms, but was adjusted based on the participant's audiovisual duration bias (see below). For non-zero conflict conditions, the visual duration was adjusted on each trial according to the conflict level (e.g., +250 ms conflict corresponded to a 750 ms nominal visual duration),

while the auditory duration remained fixed at 500 ms.

Test stimulus duration was varied across trials according to adaptive staircase procedures (1-up/3-down or 1-down/3-up in the bimodal experiment). In the bimodal experiment, we compensated for each participant's modality-specific duration bias, measured in the cross-modal task, by adjusting the visual stimulus duration. This subject-specific correction ensured that auditory and visual components were perceptually matched in the test stimuli and in the zero-conflict standard condition. Below, we describe each experiment in detail.

The duration of the visual stimulus was perceptually matched to the auditory duration (the perceptual auditory duration bias, estimated using the data from the cross-modal task, was added to the duration of the visual stimulus) in both intervals. Then the conflict duration was added to the visual duration in the standard interval. Each trial consisted of a pre-stimulus interval (200–450 ms), the first stimulus (standard or test), an ISI (400–900 ms), the second stimulus (test or standard), and a post-stimulus interval (200–450 ms), forming one continuous audiovisual stimulus.

Unimodal baselines (Auditory only, Visual Only)

In the unimodal tasks, 12 participants performed a 2IFC duration-discrimination task, in which either two auditory durations (i.e., signals embedded on a background noise) or two visual durations (i.e., visual filled disks as signals) were sequentially presented. The auditory task lasted approximately 20 minutes, and the visual task lasted around 12 minutes.

In both tasks, the standard stimulus duration was fixed at 500 ms. The duration of the test auditory stimulus varied (between 25 and 975 ms) from trial to trial. The waveform of an example stimulus in the low and high auditory-noise conditions is shown in Figure 1.

Test duration was determined using four interleaved staircases: 1-Down 2-Up, 2-Down 1-Up, 3-Down 1-Up, and 1-Down 3-Up staircases of 70 trials each, resulting in a total of 280 trials for each unimodal task. We also included 14 trials with a large duration difference between the standard and test durations (+0.45s or - 0.45s) for each task. These trials allowed us to better estimate the lapse rate (Prins, 2012).

Cross-modal duration comparison

In this task, participants compared the durations of one auditory and one visual stimulus. In each trial, the presentation order was precued. We split the experiment into 39 mini blocks of 10 trials. Before each mini block, we informed participants about the order of stimuli (e.g., “In this block the order of modalities will be: V- $\hat{A}$ ”) to orient their attention to both intervals and to understand the nature of the task. The order of stimulus presentation was randomized across mini-blocks.

Main bimodal audiovisual conflict experiment

In the main task, both the standard and test stimuli included visual and auditory durations. We introduced seven audiovisual duration conflicts, ranging from -250 to +250 ms (-250, -167, -83, 0, +83, +167, +250 ms) (Figure 5). We applied the conflict to both ends of the visual stimulus, i.e., half of the conflict magnitude extended the onset earlier and the other half extended the offset of the stimulus later. Likewise, visual PSE bias was subtracted from both ends of a visual stimulus so that a nominally zero-conflict stimulus had a visual component that was perceived as having the same duration as the auditory component. Also, note that this bias correction was applied to both the standard and test durations, while the visual-duration perturbation was only added to the standard stimulus.

Participants were asked to judge which interval had a longer duration based *only* on the auditory stimulus. Although participants were asked to judge the auditory stimulus, at the beginning of the experiment they were instructed to pay attention to both modalities and to maintain gaze at the screen even though they were not judging visual duration. We also included an audio icon to indicate that they needed to answer based on the auditory stimulus.

Each participant completed three sessions with different ranges of audiovisual conflict: (0, -167 to 250 ms), (-83 to 167 ms), and (-250 to 83 ms). The order of sessions was counter-balanced across participants. As in the unimodal condition, easy lapse-rate trials were included in each experimental block (14 trials per condition; noise level $\times$ conflict). For each condition, participants completed four interleaved staircases (two 1-down-2-up and two 2-down-1-up), each consisting of 70 trials. Across all conditions and sessions, this resulted in a total of 2,156 trials for the main task.

Staircases

In all tasks, we measured discrimination thresholds using adaptive staircases, as described in each task section. Step-size reduction was applied after each staircase reversal. Specifically, whenever the staircase changed direction (from increasing to decreasing test duration, or vice versa), the step size was halved. Step-size reduction was applied only for the first two reversals; subsequent steps remained fixed.

The initial step size in the staircase procedure was defined as the duration difference percentage (Δ) between the test and standard durations. At the beginning of the experiment, Δ was set to the largest negative value allowed, corresponding to -95% of the standard duration. Since the standard duration was 0.5 s, this yielded an initial test interval of $0.5 - 0.95 \times 0.5 = 0.025$ s. This value represents the shortest possible test interval used in the experiment.

The starting value of the staircase (Δ) was set such that the resulting change in duration corresponded to approximately 25 ms for both auditory and visual cues. This ensured that the initial difference was clearly perceptible for participants.

Results

Cue reliability

To measure sensory precision, all participants completed two unimodal duration-discrimination tasks: one auditory and one visual (Figure 1; see Methods for details). For each task, we fit log-normal psychometric functions to individual participant data using maximum-likelihood estimation.

The log-normal observer model is well suited for duration judgments, as it naturally captures scalar variability consistent with Weber’s law and constrains duration estimates to be positive (Haigh et al., 2021). The psychometric function included two parameters: a lapse rate (λ), capturing stimulus-independent response errors, and a sensitivity parameter (σ_v for the visual task and $\sigma_{a,l}$ and $\sigma_{a,h}$ for the low and high auditory-noise conditions), reflecting the uncertainty in sensory encoding (steepness of the psychometric function) in log-duration space.

We did not include a bias parameter (μ) in the main analyses. In a symmetric two-interval, forced-choice (2IFC) design, any additive bias in perceived log-duration would affect both the standard and test intervals equally and therefore cancel in the com-

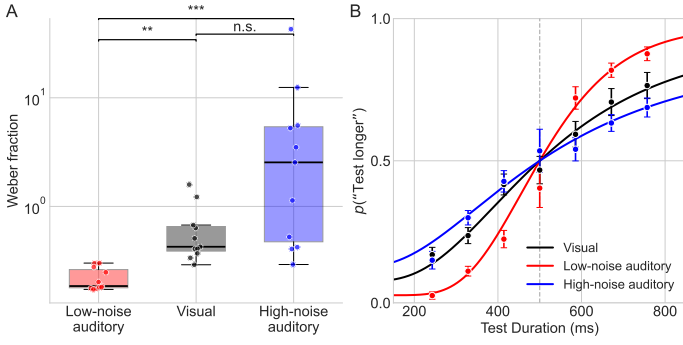


Figure 2: Unimodal visual and auditory duration-discrimination psychometric functions. **A** Unimodal discrimination thresholds expressed as Weber fractions, derived from log-normal psychometric fits, for the low-noise auditory (red), visual (black), and high-noise auditory (blue) conditions. Each point represents an individual participant. Boxplots display the interquartile range (IQR; 25th–75th percentile), with the median indicated by the central line. Whiskers extend to the most extreme values within $1.5 \times \text{IQR}$ of the first and third quartiles; values beyond this range are treated as outliers. Data points are horizontally jittered for visualization clarity. **B** Psychometric functions showing $p(\text{“test longer”})$ as a function of test duration, pooled across all participants and trials. Data points represent proportions calculated within duration bins; error bars indicate binomial standard errors. Solid lines show fitted log-normal psychometric functions. Red: low-noise auditory; black: visual; blue: high-noise auditory. Discrimination precision was highest for low-noise auditory, intermediate for visual, and lowest for high-noise auditory conditions.

parison. Consequently, when the standard (t_s) and the test (t_t) durations are identical, the model predicts $P(\text{“test longer”}) = 0.5$ by symmetry, and no bias term is theoretically required.

Under this model, the probability of judging the test interval as longer than the standard is given by

$$P(\text{“test longer”}) = \frac{\lambda}{2} + (1 - \lambda) \Phi\left(\frac{\log(t_t) - \log(t_s)}{\sigma}\right), \quad (1)$$

where t_t and t_s denote the test and standard durations, respectively, and Φ is the standard normal cumulative distribution function. Figure 2 shows the unimodal duration-discrimination thresholds (Weber fraction; Panel A) and the psychometric-function fits (Panel B) for the auditory (high/low noise) and visual conditions.

The Friedman test and Wilcoxon signed-rank test analysis revealed significant effects of the auditory noise level on temporal-discrimination precision. We conducted all analyses on the raw log-space noise estimates (σ_v , $\sigma_{a,l}$ and $\sigma_{a,h}$). However, for reporting purposes we converted these to just noticeable differences

(JNDs) in milliseconds, defined as the minimum duration increment required to achieve 75% correct performance and then to Weber fractions (WF). Given a fitted value of σ , the JND at 75% accuracy is:

$$\text{JND} = t_s (\exp(z_{75}\sigma) - 1), \quad (2)$$

where t_s is the standard duration and $z_{75} = \Phi^{-1}(0.75)$ is the inverse cumulative normal value corresponding to 75% correct.

We then computed the WF as the proportional threshold:

$$\text{WF} = \frac{\text{JND}}{t_s}. \quad (3)$$

For the low- and high-noise auditory and visual conditions, the estimated Weber fractions were 0.18, 2.54, 0.42, respectively, with corresponding median JNDs of 92.5, 1271.8, and 212.7 ms. A Friedman test revealed a significant difference across conditions ($\chi^2(2) = 18.73, p < .001$; Figure 2), indicating that discrimination thresholds differed across the auditory (low- and high-noise) and visual conditions.

Pairwise comparisons were conducted using Wilcoxon signed-rank tests with Bonferroni correction ($\alpha = .0167$). $\sigma_{a,l}$ (low auditory-noise condition) was significantly lower than $\sigma_{a,h}$ (high auditory-noise condition; $W = 0, p < .001$). The value $W = 0$ indicates that, for every participant, the paired difference $\sigma_{a,h} - \sigma_{a,l}$ was positive, i.e., $\sigma_{a,l}$ was lower than $\sigma_{a,h}$ within all participants. Moreover, $\sigma_{a,l}$ was significantly lower than σ_v ($p < .001$). The difference between σ_v and $\sigma_{a,h}$ was not significant after Bonferroni correction ($p = 0.18$).

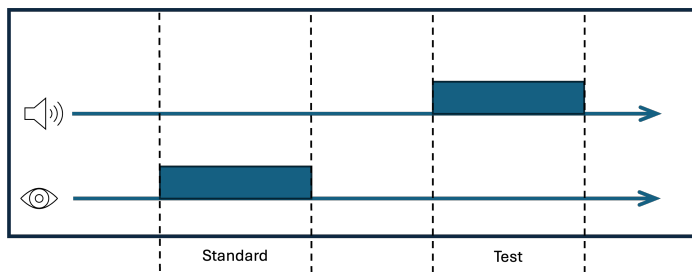


Figure 3: **Cross-modal duration-comparison trial structure.** Schematic of a single cross-modal trial. The trial consisted of two sequential stimuli (standard and test), separated by an inter-stimulus interval (dashed vertical lines indicate interval boundaries). **Top (auditory stream):** Continuous background noise was presented throughout the trial. A signal burst occurred during either the first or second interval. **Bottom (visual stream):** A disk outline was continuously visible. During the signal duration, the disk was filled. In this task, the visual stimulus was the standard, with a duration of 500-ms. The auditory test duration varied across trials. The auditory test and visual standard were presented in random order.

Modality-specific duration biases

To estimate participants' inherent modality-specific biases in duration perception, we included a cross-modal duration-comparison task (Figure 3). In this experiment, participants were presented with an auditory signal (embedded in continuous auditory noise) and a visual signal (a filled disk against an otherwise empty background). The presentation order of these stimuli was precued at the beginning of each mini-block of 10 trials. Participants reported which stimulus had the longer duration.

Psychometric fits revealed systematic biases in cross-modal judgments, with bias magnitude depending on auditory noise level (Figure 4). Under the low auditory-noise condition, the median point of subjective equality (PSE) shift was -116 ms, indicating that the auditory duration had to be physically shorter than the visual duration to be perceived as equal. A Wilcoxon signed-rank test confirmed that this shift differed significantly from zero ($W = 6, p = .014$). Under the high auditory-noise condition, the median PSE shift was -73 ms, indicating a reduced auditory bias. However, this shift did not differ significantly from zero. Together, these results indicate a reliable overestimation of auditory relative to visual duration in the low-noise condition, with a weaker bias under high auditory noise.

Given the systematic modality-specific biases observed in the cross-modal task, we estimated each participant's subjective

auditory–visual duration bias and compensated for it in the bi-modal experiment. Specifically, we adjusted the visual duration by adding half of the measured bias to the onset and half to the offset of the visual duration. This procedure preserved audiovisual synchrony while ensuring that, in the absence of added conflict, auditory and visual components were perceptually matched in duration.

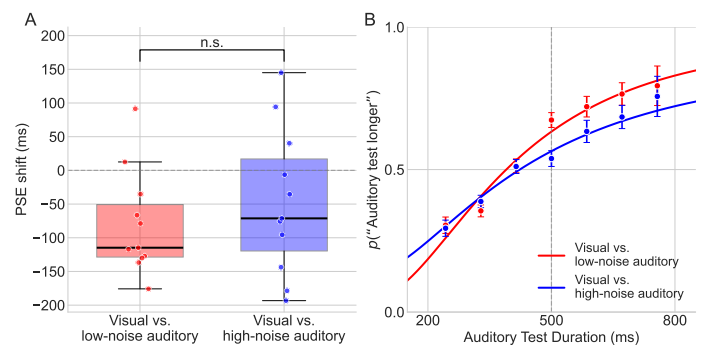


Figure 4: **PSE shifts and psychometric curve fits for the cross-modal duration comparison.** **A** Point of subjective equality (PSE) shifts for the low-noise auditory (red) and high-noise auditory (blue) conditions. Superimposed are the individual participant data points (horizontally jittered). The PSE shift is the difference between the standard duration (500 ms) and the auditory duration that is perceived as identical to the standard. **B** Psychometric functions (dots) and model fits (solid lines), showing p (“auditory test longer”). Binary data were grouped into 7 equally-spaced bins between 200 and 800 ms (error bars: SEM). For this plot we pooled the data across all participants then fitted with log-normal cumulative distribution functions (parameters: λ : lapse rate; μ : PSE, σ : log-space noise). Solid lines show fit log-normal psychometric functions.

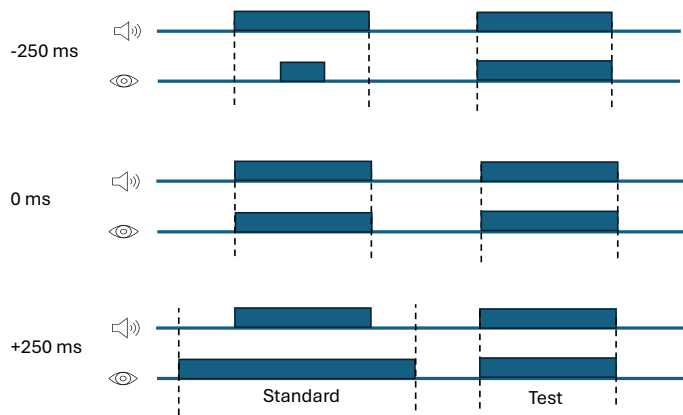


Figure 5: **Audiovisual conflict in the bimodal experimental task.** Conflicts were introduced exclusively in the visual component of the *standard* stimulus, while the auditory component remained unchanged. The top row shows an example negative-conflict (-250 ms), in which the visual duration of the standard stimulus was shorter than the auditory duration. Across trials, audiovisual conflict was randomly sampled from seven levels ranging from -250 ms to $+250$ ms. Note that the visual durations in both the standard and test stimuli were also corrected for the participant's audiovisual bias as estimated from the cross-modal task.

Main experiment: Cue-conflict effects

In the main bimodal experiment, participants were asked to judge which of two stimulus intervals was longer based solely on the auditory duration (Figure 5). Despite this instruction, participants' judgments were systematically influenced by the duration of the visual stimulus, especially under high auditory noise.

To investigate how participants combined auditory and visual cues during duration judgments, we analyzed psychometric functions across varying levels of audiovisual conflict and auditory cue reliability. Figure 6 shows the psychometric functions for each conflict level, separately for the low (left) and high (right) auditory-noise conditions. As the visual duration increased, the curves shifted horizontally toward longer durations, indicating that the task-irrelevant visual cue biased auditory duration judgments. This bias was more pronounced under the high-noise condition, qualitatively consistent with the predictions from reliability-weighted cue combination. Under low noise, psychometric functions showed smaller shifts, reflecting a stronger reliance on the auditory modality.

To quantify the results, we extracted the PSE from each psychometric curve and plotted it as a function of the conflict level (Figure 7). Under the high auditory-noise condition, PSEs

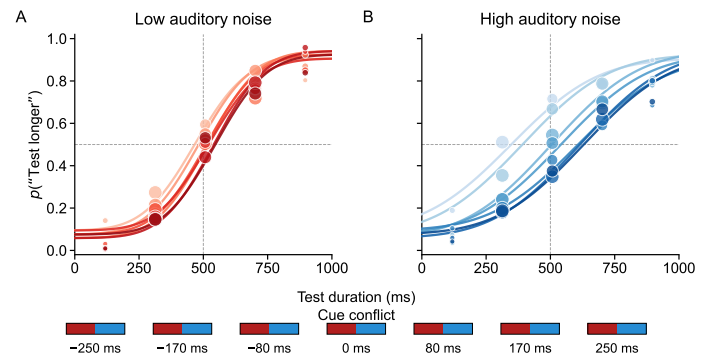


Figure 6: **Data and fitted psychometric functions in the bimodal experiment.** **A** Fitted psychometric functions in the low auditory-noise condition. Each curve corresponds to one of seven audiovisual conflict levels (-250 to $+250$ ms). Dots show p (“test longer”) as a function of seven equally spaced test-duration bins. Data points are plotted at the centers of the bins, with dot diameter proportional to the number of trials within each bin. **B** Same as **A**, but for the high auditory-noise condition. Psychometric curves shift progressively with conflict magnitude, indicating stronger visual influence on perceived auditory duration under high auditory noise.

showed a nearly linear relationship with conflict magnitude. In contrast, under the low-noise condition, PSEs stayed close to zero, indicating reduced bias toward visual durations. The error bars reflect the 95% confidence interval from 1000 parametric bootstrap samples. For each sample, we generated responses from the fitted psychometric function, refit the function, and computed the standard error of the resulting parameter estimates. Regression analysis confirmed a linear relationship between cue-conflict magnitude and PSE shifts. Under the high-noise condition, visual conflict magnitude showed a strong positive correlation with PSE shifts ($r = 0.963$, $p < 0.001$), with a steep slope of 0.606 ms PSE change per 1 ms of visual conflict. In contrast, under the low-noise condition, the correlation remained strong ($r = 0.911$, $p = 0.004$), but had a much shallower slope of 0.144 ms per ms conflict. These results indicate that auditory duration perception was systematically biased toward the visual duration, and that the strength of this bias was modulated by auditory cue reliability.

Computational modeling

To explain participants' behavior in the audiovisual duration-conflict task, we fit five models to the behavioral data. These models shared the same encoding process but differed in decod-

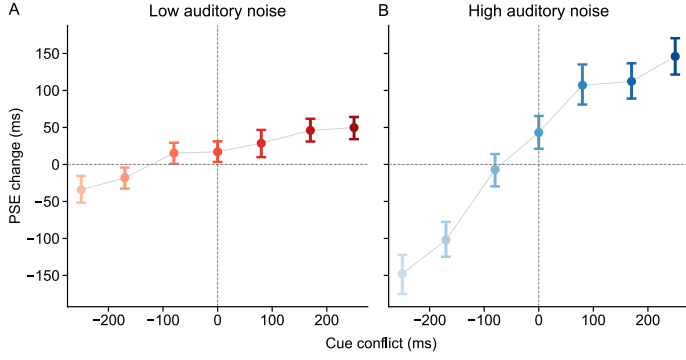


Figure 7: **Effect of audiovisual conflict on perceived auditory duration.** Group-averaged point of subjective equality (PSE) estimates as a function of audiovisual duration conflict, shown separately for the low-noise auditory condition (A) and high-noise auditory condition (B). Each data point reflects the mean PSE of pooled data over 11 participants; error bars show bootstrapped 95% confidence intervals. Under high noise, auditory duration judgments are pulled toward the visual duration. Under low noise, PSEs are more stable across conflict levels, suggesting auditory dominance.

ing strategy. Below, we describe the models' encoding process, followed by the decoding stage.

Sensory encoding

All models assumed that internal representations of auditory and visual durations are noisy. To capture Weber-like scaling of temporal uncertainty (scalar variability), we assumed that durations are encoded in logarithmic space with additive Gaussian noise:

$$p(m_a | s_a) \sim \mathcal{N}(\log s_a, \sigma_a^2), \quad p(m_v | s_v) \sim \mathcal{N}(\log s_v, \sigma_v^2). \quad (4)$$

Here, s_a and s_v denote stimulus durations in linear time units, whereas m_a and m_v represent internal measurements in log-duration space. Because encoding noise was Gaussian in log space, the implied distribution of internal measurements expressed in linear time is log-normal. This formulation naturally produced scalar variability, such that uncertainty scaled proportionally with stimulus duration. Scalar variability is a characteristic feature of perception of time and motivates this log-duration encoding. Thus, both encoding and decoding were performed in log-duration space (Killeen and Weiss, 1987; Wittmann, 2009; Ren et al., 2020).

Decoding strategy

While all tested models shared the same encoding process, they differed in how encoded information is integrated and read out across sensory modalities. We considered five classes of integration and decision strategies: forced fusion, three forms of causal inference, and heuristic cue selection. We describe each of the five models next.

Forced fusion (Optimal cue integration)

The forced-fusion model assumed that observers integrate auditory and visual signals for the auditory-duration judgment regardless of cue-conflict magnitude. Moreover, the integration is optimal, with each cue weighted by its reliability (Eq. 32).

This model can be viewed as a special case of the Bayesian causal-inference model in which the prior probability of common cause is fixed at $p_c = 1$, i.e., the two cues are always integrated regardless of cue-conflict magnitude. Deviations from this model provide evidence that integration strength varies as a function of cue conflict and uncertainty.

Causal-inference models

The Bayesian causal-inference model assumed that the observer first infers whether the auditory and visual cues arise from a common source by computing the posterior probability of a common source given the two cues' sensory measurements:

$$p(C = 1 | m_a, m_v), \quad (5)$$

where $C = 1$ denotes a common cause. The observer then forms a posterior over the task-relevant variable (here, the auditory stimulus duration s_a) by averaging over two posteriors. The inferred output is the estimate $\hat{s}_a$:

$$p(s_a | m_a, m_v) = p(C = 1 | m_a, m_v) p(s_a | m_a, m_v, C = 1) + p(C = 2 | m_a, m_v) p(s_a | m_a, C = 2), \quad (6)$$

in which $p(C = 2 | m_v, m_a)$ indicates the posterior probability of separate causes. $p(s_a | m_v, m_a, C = 1)$ and $p(s_a | m_a, C = 2)$ specify the posteriors over s_a under each causal scenario. The posterior probability of common cause was computed as:

$$p(C = 1 | m_a, m_v) = \frac{p(m_a, m_v | C = 1) p_c}{p(m_a, m_v | C = 1) p_c + p(m_a, m_v | C = 2) (1 - p_c)}. \quad (7)$$

We tested three causal-inference models, in which the posterior probability of common cause $p(C = 1 | m_a, m_v)$ was computed in the same way. However, the models differed in how this posterior probability was applied to form the final duration estimate used for the decision. Specifically, the models used the posterior probability to (i) average the estimates under the common- and separate-cause hypotheses weighted by their posterior probabilities, (ii) select the estimate associated with the more probable causal structure, or (iii) stochastically sample between these estimates according to the posterior probability. Each model included a prior probability of common cause, p_c , which was fitted to the data.

Likelihoods under the common-cause and separate-causes hypotheses were computed by marginalizing over the latent true stimulus durations using a bounded prior, following the standard formulation of Bayesian causal inference (Körding et al., 2007). Full expressions, choice of prior distribution, and analytical derivations are provided in [Appendix](#), specifically in the [Bayesian causal-inference model subsection](#).

We modeled the prior distribution of durations as a truncated uniform distribution over the range of stimulus durations presented in the experiment, $[t_{\min}, t_{\max}]$, providing finite support for the latent duration variable.

Model averaging

This model assumed that participants inferred the stimulus duration based on the inferred causal relationship between the cues presented in the trial. It assumed that the observer started by inferring the posterior probability of both cues originating from a common cause by computing $p(C = 1 | m_a, m_v)$. Under this model, observers computed two estimates of the auditory stimulus duration, one based on both auditory and visual measurements, and another based on the auditory measurement alone. The final auditory duration estimate was a weighted average of these two separate estimates, given by:

$$\hat{S}_a = p(C = 1 | m_a, m_v) \hat{S}_{a,C=1} + (1 - p(C = 1 | m_a, m_v)) \hat{S}_{a,C=2}, \quad (8)$$

where $\hat{S}_{a,C=1}$ and $\hat{S}_{a,C=2}$ denote the estimates of the auditory stimulus under the common-cause and independent-causes assumptions, respectively. This strategy produced an estimate that smoothly interpolated between fusion and segregation as a function of sensory conflict and uncertainty, resulting in graded biases toward the more reliable cue. Model averaging is performed in log-duration space, consistent with the assumed log-Gaussian sensory encoding and inference framework.

Model selection

The model-selection strategy treats the causal structure as a categorical latent variable and assumes that observers committed to a single causal interpretation on each trial. Specifically, instead of averaging estimates across the two causal hypotheses, observers selected the estimate derived from the more probable causal structure (i.e., the one with higher posterior probability).

In our implementation, when the posterior probability of common cause exceeded 0.5, observers fully integrated auditory and visual measurements to form a fused estimate. Otherwise, observers assumed separate causes and based their judgment solely on the auditory measurement, consistent with task instructions. This strategy prioritizes categorical causal decisions and predicts a sharper transition between integration and segregation as a function of sensory conflict compared to model averaging.

Probability matching

The probability-matching strategy preserves the uncertainty about the causal structure at the level of individual trials. Rather than committing to a single causal interpretation or continuously averaging across hypotheses, observers stochastically sampled causal structures in proportion to their posterior probability. Although probability matching is computationally simple and widely observed in human decision-making (Wozny et al., 2010; Ren et al., 2020), it is suboptimal for both estimation accuracy and causal classification.

Under this strategy, fusion is selected with probability $p(C = 1 | m_a, m_v)$, while segregation is selected with probability $1 - p(C = 1 | m_a, m_v)$. When segregation is selected, judgments are based solely on the auditory cue measurement, consistent with task demands. Probability matching produces trial-to-trial variability in integration behavior and yields mixture-like response patterns without continuous averaging of the common-cause and

separate-causes estimates.

Probabilistic cue switching

We also considered a probabilistic cue-switching model, in which observers based their decision on either the auditory or the visual cue on each trial (see Methods). Instead of computing posterior probabilities of causal structure, this heuristic strategy selected a single cue stochastically: with fixed selection probability p_v , the decision was based solely on the visual estimate, and with probability $1 - p_v$, it was based solely on the auditory estimate. Thus, p_v represents the probability of relying on the visual cue on a given trial. This formulation captures the possibility that observers rely on stable response tendencies rather than dynamically weighting sensory information. Despite its simplicity, the switching strategy can reproduce cue-integration-like patterns when responses are averaged across trials.

Modeling results

We first evaluate how well the models accounted for the shifts in PSE as a function of audiovisual cue conflict across the two auditory noise levels. We compared observed PSE shifts with predictions from five models: forced fusion, model averaging, model selection, probability matching and probabilistic cue switching. Sensory-noise parameters were fitted using the data from the bimodal experiment.

Figure 8 shows model fits to the PSE-shift data. For ease of visualization, predictions of only three of the models are displayed: forced fusion, model averaging and probabilistic cue switching. All three models capture the overall pattern that perceived auditory duration was systematically biased toward visual duration and that the bias increased approximately linearly with audiovisual conflict magnitude. All three models also predict an overall larger bias under higher auditory noise. This convergence suggests that our data may be limited in the ability to adjudicate between these models. A comparison that includes all five models and fits to individual participants' data is presented in the Appendix (Figure 11).

Causal-inference models typically predict a non-linear breakdown of integration for large cue conflicts. Our data show only weak deviations from linearity. These deviations are more pronounced for large positive conflicts than for large negative conflicts, indicating an asymmetry in bias magnitude across the con-

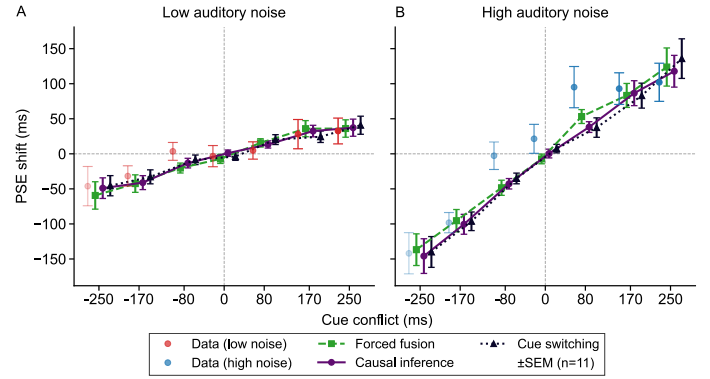


Figure 8: Conflict-dependent biases in auditory duration perception. Point of subjective equality (PSE) shifts as a function of audiovisual conflict for the low auditory-noise condition (A) and high auditory-noise condition (B), averaged over 11 participants. Positive values indicate biases toward longer perceived auditory duration. Red and blue show empirical data (group mean $\pm$ SEM across 11 participants). Other lines indicate model predictions for forced fusion, causal model averaging, and probabilistic cue switching. The model predictions are jittered horizontally for readability. All model predictions were generated using sensory-noise parameters fitted using the bimodal data. For models, error bars reflect $\pm$ SEM across participant fits.

flict range. This pattern suggests that, at the group level, observers do not exhibit the strong nonlinearity predicted by causal-inference models in which cue integration turns to cue segregation with increases in cue conflict. Instead, integration remains approximately linear within the tested conflict range. As a result, the behavioral signatures of causal inference are weak, limiting our ability to discriminate between causal-inference models and simpler integration or heuristic strategies with these data.

Model fitting and model comparison

All models have a lapse-rate (λ) parameter, two auditory duration-noise parameters (one for each auditory-noise condition: $\sigma_{a,l}$ and $\sigma_{a,h}$), and one visual duration-noise parameter (σ_v). For models that incorporate causal inference, there is an additional parameter (p_c), which captures the prior probability of a common source. For the probabilistic cue-switching model, there are two additional parameters ($p_{v,l}$ and $p_{v,h}$) that specify the probability of switching (i.e., relying on vision rather than audition) in each of the two auditory-noise conditions.

We fit each model to each participant's data (from the two auditory-noise conditions jointly) using maximum-likelihood es-

timation. For each model, we computed the predicted probability of responding “test longer” for every stimulus condition via Monte Carlo simulation. Specifically, on each simulated trial, we drew noisy internal measurements of auditory and visual durations from the encoding model, applied the model’s integration or decision rule to produce an estimate of the auditory stimulus, and then determined whether the test estimate was longer than the standard estimate. For each condition, that is, for each combination of auditory test duration, standard duration, cue conflict, and noise level, we simulated 2000 Monte Carlo trials (i.e., repeated draws of internal sensory measurements). For each Monte Carlo trial we generated samples of the auditory and visual internal measurements according to the model’s encoding-noise assumptions and determined the model’s response for that trial. The predicted response probability for each condition was computed as the proportion of simulated trials for that condition for which the model observer judged the test duration as longer. The log-likelihood of each model was then computed using the binomial likelihood of the observed responses given the predicted probabilities across all conditions. Parameter bounds were chosen based on plausible ranges for sensory noise and lapse rates. In causal-inference models, the prior over stimulus duration was assumed to be uniformly distributed over the range of test durations presented for each participant. Parameters were estimated by minimizing the negative log-likelihood using Bayesian Adaptive Direct Search (Acerbi and Ma, 2017). To reduce the risk of local minima, the optimization was repeated from 10 random initial parameter values.

Sensory-noise parameters were fitted using the data from the bimodal task, because effective noise may differ between unimodal and bimodal contexts due to attentional allocation and cross-modal interactions (Badde et al., 2020). To ensure robustness, we also evaluated all models using sensory-noise parameters estimated using the psychometric function fits to the unimodal-task data. The results using these parameter estimates were qualitatively consistent (Figure 12).

We evaluated model performance using the Akaike Information Criterion (AIC) at the individual participant level. Model comparison was performed under two assumptions about sensory noise: (a) sensory noise fixed to values estimated from unimodal task data and (b) sensory noise fitted directly using the bimodal data (Figure 9).

Consistent with the observation that all models captured the data reasonably well (Figure 8), model comparison revealed only small differences in fit quality across integration strategies. Below, we report model-comparison results under two conditions: sensory-noise parameters were (1) fixed to values estimated from the unimodal experiment, or (2) treated as free parameters and fit directly to the bimodal data.

Fixed sensory noise

In this analysis, we assume that sensory noise is identical for unimodal and bimodal stimuli. Thus, we estimated the noise parameters using the data from the unimodal duration-discrimination tasks, and then used those estimates when fitting the data from the bimodal task. Figure 9A shows the participant-level ΔAIC values, computed as the difference between the AIC of the best-fitting model for each participant and the AIC values of the other models. For most participants (8 out of 11), the probabilistic cue-switching model achieved the lowest ΔAIC among the candidate models, indicating that it provided the best trade-off between goodness of fit and model complexity. The Bayesian causal-inference variants and the forced-fusion model had larger ΔAIC values. The forced-fusion model was the best-fitting model for only three participants. No other model achieved the lowest ΔAIC for any participant.

At the group level, the summed ΔAIC values across participants yielded the same pattern, with the probabilistic cue-switching model overall performing the best (Figure 9A).

Free sensory noise

When sensory-noise parameters were fitted using the bimodal data, model comparison revealed substantially weaker separation between candidate models (Figure 9B). Participant-level ΔAIC values showed considerable overlap between models, indicating that multiple integration strategies provided similarly good fits once sensory noise was allowed to vary.

The probabilistic cue-switching model provided the best fits for the largest number of participants (4 out of 11; Figure 9B). At the group level, it also yielded the lowest summed ΔAIC across participants. However, the individual ΔAIC values were very low, suggesting limited discriminability between integration strategies under this assumption.

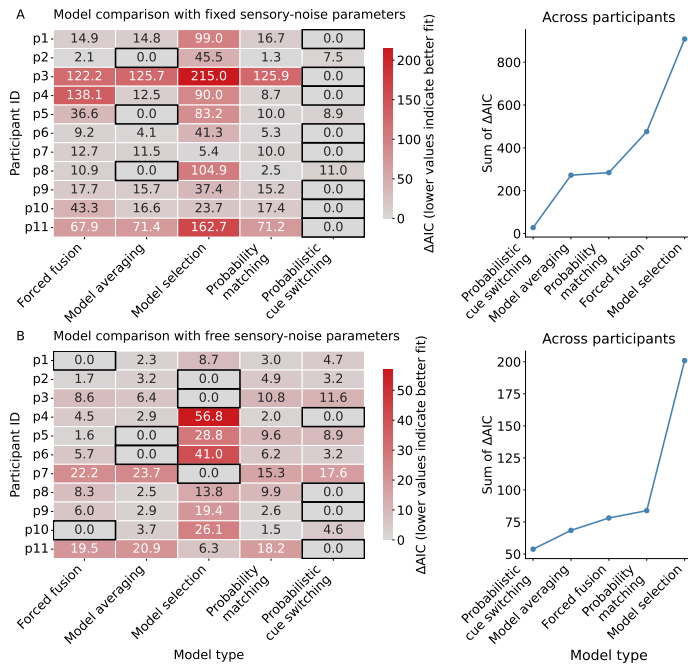


Figure 9: **A** Model comparison assuming sensory-noise parameters fixed to values estimated from the unimodal tasks. **B** Model comparison when sensory-noise parameters were additionally fit to the bimodal data. Left panels show participant-level ΔAIC values for each model (lower values indicate better fit), computed relative to the best-fitting model for that participant. Rows correspond to participants and columns correspond to models. Right panels show the sum of ΔAIC values across participants for each model, providing a group-level comparison. When sensory-noise parameters were fixed (A), the probabilistic cue-switching model provided the best fit for 8 of 11 participants. When the sensory-noise parameters were fit directly to the bimodal data (B), the cue-switching model provided the best fit for 4 of 11 participants.

Random-effects Bayesian model selection

To assess model distinguishability at the population level, we performed random-effects Bayesian model selection (Stephan et al., 2009; Rigoux et al., 2014). This approach does not assume that all participants use the same model. Instead, it estimates the distribution of models across the population and quantifies the uncertainty about which model is most prevalent. Under this framework, we first computed the exceedance probability (XP), defined as the posterior probability that a given model is more frequent in the population than all competing models. Although the probabilistic cue-switching model yielded the highest raw exceedance probability ($\text{XP} = 0.52$), this value alone does not necessarily account for the possibility that the observed differences in model evidence across participants might have been due to chance rather than differences in population-level model prevalence. To address this, we calculated the protected exceedance probability (PXP) (Rigoux et al., 2014), which adjusts the exceedance probability by incorporating the Bayesian Omnibus Risk (BOR), defined as the posterior probability that the observed differences in model evidence across participants could arise under a null hypothesis in which all models are equally frequent in the population. The Bayesian Omnibus Risk was high ($\text{BOR} = 0.88$), indicating a substantial probability that the observed differences in model evidence reflect chance fluctuations rather than true population-level differences in model prevalence. After this correction, protected exceedance probabilities were approximately uniform across models (all $\text{PXP} \approx 0.18\text{--}0.24$), with the cue-switching model yielding $\text{PXP} = 0.24$, only slightly above chance level ($1/5 = 0.20$). None of the models approached the conventional threshold for strong evidence ($\text{PXP} \geq 0.95$). Together, these results indicate that no model clearly predominates at the population level.

High sensory-encoding noise limits model discriminability

To explain the lack of evidence for causal inference even at large audiovisual cue conflicts, we recovered the posterior probability of common cause $P(C = 1 \mid m_a, m_v)$ across conflict levels using each participant's fit parameters.

Figure 10 shows $P(C = 1 \mid m_a, m_v)$ for an example participant. Across all conflict levels and auditory-noise conditions, the posterior probability of common cause remained relatively high (i.e., above 0.5), a pattern that was maintained for all participants

(Figure 13). This is due to the large sensory encoding noises estimated from the data. Specifically, when sensory encoding noise is high, even substantial physical conflicts produce overlapping measurement distributions, yielding consistently high posterior probability of common cause. Because the two log likelihoods decrease at comparable rates as conflict grows, their ratio remains approximately constant, and the posterior stays well above 0.5 across all conflict levels especially in the high auditory-noise condition. This indicates that, for this participant, the observer’s fitted parameters are consistent with treating the auditory and visual signals as originating from a common source regardless of the imposed conflict, explaining why the causal-inference and forced-fusion models produce similar predictions.

This analysis reveals why causal-inference models and forced-fusion models produced similar predictions in our experiment. Given the estimated sensory-noise parameters, the posterior probability of common cause never dropped low enough to produce substantial segregation of cues. The observer’s inference remained dominated by the common-cause hypothesis across the entire conflict range, resulting in near-linear, fusion-like behavior. Within the conflict range examined here, competing models predict largely overlapping, noisy psychometric patterns, imposing a fundamental limit on model identifiability rather than reflecting insufficient data or model distinguishability.

Discussion

In this study, we investigated how humans infer duration under conflicting audio and visual cues with varying levels of sensory uncertainty. Previous work has shown that audiovisual duration cues can be integrated in a statistically optimal, reliability-weighted manner (Shams et al., 2005; Hartcher-O’Brien et al., 2014). Extending this framework, we asked whether such integration is flexibly modulated by causal inference, as observed in audiovisual spatial-localization tasks where large cue conflicts induce non-linear breakdowns of integration (Körding et al., 2007; Badde et al., 2021; Hong et al., 2021). Contrary to this expectation, we found little evidence for a pronounced non-linear reduction in integration strength. Instead, auditory duration judgments remained largely linear and noise-dependent even at relatively large conflicts, suggesting that short sub-second duration judgments may rely more strongly on heuristic integration strategies than on explicit inference about causal structure.

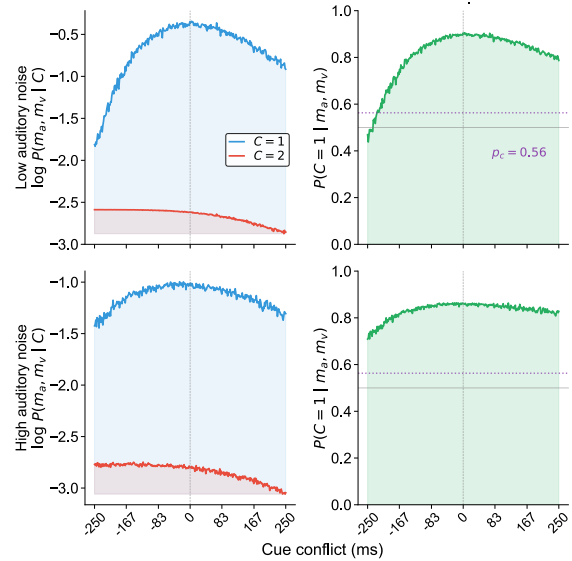


Figure 10: **Posterior probability of common cause across conflict levels.** Log likelihoods and posterior probability of common cause for an example participant, plotted as a function of audiovisual duration conflict. Top row: low-auditory-noise condition; bottom row: high-auditory-noise condition. Left column: log likelihood of the measurements under a common cause, $\log P(m_a, m_v | C=1)$ (blue) and under separate causes, $\log P(m_a, m_v | C=2)$ (red), plotted on the same axes so that their relative magnitudes can be compared directly. Right column: Posterior probability of common cause, $P(C=1 | m_a, m_v)$, with the dotted line marking the fitted prior p_c . All quantities were computed via Monte Carlo simulation (500 samples per conflict level) using participant-specific sensory-noise estimates ($\sigma_{a,l}$, $\sigma_{a,h}$, σ_v) and common-cause prior (p_c) obtained from the log-normal causal-inference model fit.

To interpret these findings, we compared a set of models spanning Bayesian causal inference and probabilistic cue-switching strategies. Although a probabilistic cue-switching model provided the best overall fit under several model-comparison criteria, forced-fusion and causal-inference models produced highly similar predictions within the experimentally plausible conflict range. While these candidate models differ substantially in their computational assumptions, ranging from explicit causal inference to probabilistic cue switching, their behavioral predictions converge under the levels of sensory conflict and uncertainty tested here. This convergence highlights a fundamental limitation of model identifiability. Within the conflict range tested here (± 250 ms), and in the absence of externally added visual noise (i.e., visual stimuli with clear onset and offset), effective sensory uncertainty remains substantial for at least the visual modality (e.g., $\sigma_v = 0.51$, corresponding to a JND of approximately 200 ms). With these conditions, behavioral biases are close to linear functions of audiovisual conflict. As a result, these distinct integration strategies result in highly similar duration-discrimination patterns, making them difficult to distinguish based on behavior alone.

Consistent with this interpretation, the bimodal auditory duration judgments were less variable than predicted from the unimodal auditory and unimodal visual performance, and in some cases even less variable than the optimal-integration predictions based on unimodal noise estimates. We speculate that visual signals in the bimodal task contributed not only to duration information but also to interval segmentation. However, in unimodal auditory conditions, particularly under high noise, very short intervals may be difficult to detect, making it harder to segment the two intervals reliably, leading to increased response variability. Visual input may therefore stabilize auditory duration judgments beyond what would be expected from cue integration alone.

Causal inference versus heuristic strategies

In the present study, both Bayesian causal-inference models and heuristic strategies were able to capture the main behavioral regularities observed in the auditory duration judgments, including noise-dependent biases toward visual duration. Notably, the probabilistic cue-switching model provided equal or superior fits to the behavioral data compared to causal-inference variants across multiple model-comparison criteria. At the same time,

causal-inference models did not produce uniquely diagnostic predictions within the experimentally tested conflict range. This convergence suggests that explicit inference about the causal structure does not contribute to the observed behavior.

The lack of evidence for causal inference should not be interpreted as evidence against its plausibility. Rather, our results highlight a fundamental limitation in discriminating between integration strategies when sensory noise is sufficiently high. Under these conditions, the posterior probability of separate causes remains low even at large cue conflicts. As a result, bias toward the visual stimulus continues to increase approximately linearly with increasing cue conflict. In this regime, heuristic strategies that stochastically select between cues can closely approximate the predictions of more complex inferential models. Recent work has shown that probabilistic switching strategies can reproduce behavioral patterns often interpreted as Bayesian integration, even in the absence of explicit inference over causal structure (Ni and Ma, 2024). This perspective aligns with our finding that multiple integration strategies produce near-indistinguishable behavioral signatures in audiovisual-duration perception.

Limits of identifiability and factors shaping audiovisual duration perception

The limited identifiability shown here is not specific to the present dataset or modeling choices, but reflects a broader constraint on inferring computational mechanisms from behavioral data in temporal tasks. When multiple models yield near-equivalent predictions over the observable stimulus range, model comparison becomes sensitive to minor differences in parameterization rather than to principled differences in integration strategy. As a result, evidence for causal inference based solely on behavioral fits should be interpreted with caution in duration-discrimination paradigms.

Several task-specific factors further constrain the observable integration behavior. Perceived duration differs systematically across modalities even when physical durations are matched, with auditory intervals typically perceived as longer than visual intervals. To prevent such baseline biases from influencing the measured audiovisual bias, we compensated for modality-specific differences before introducing audiovisual conflict. Under these controlled conditions, auditory duration judgments showed robust, noise-dependent biases toward visual duration, confirming

that visual influence scales with relative sensory reliability even when synchrony and baseline perceptual differences are controlled.

Furthermore, because the task in the main bimodal experiment was to judge the auditory duration component of the audiovisual stimulus, it does not allow a direct assessment of whether integration strategies are symmetric. However, because the experiment included two auditory-noise conditions and the threshold of discriminating visual durations fell between those of the two auditory conditions, we would expect to observe similar results if participants were instead asked to judge the duration of the visual component of the audiovisual stimulus.

Together, these considerations show that audiovisual duration integration is jointly constrained by sensory uncertainty, modality-specific biases, and fundamental limits on model discriminability within perceptually plausible conflict ranges.

Asymmetry between shorter and longer audiovisual bias

Although most participants exhibited approximately linear bias patterns as a function of audiovisual conflict, the aggregated data revealed modest deviations from linearity at larger conflicts, particularly for conflicts in which visual duration exceeded auditory duration. These asymmetries were weaker or absent for conflicts in which the auditory duration was longer and did not produce the pronounced non-linear breakdown predicted by classical causal-inference models.

The observed asymmetry in bias magnitude across opposing audiovisual conflicts was not anticipated by our initial hypotheses. Under a log-space encoding assumption, shorter durations are expected to be represented with lower uncertainty, which should lead to a stronger breakdown of integration for conflicts with a briefer stimulus. Contrary to this prediction, deviations from linearity were more apparent for large conflicts with a longer-duration visual stimulus.

One possible explanation is that factors beyond duration integration may influence performance at extreme conflicts. In particular, very short visual intervals may be less salient or less informative as temporal markers, reducing their influence on auditory judgments. Because this pattern was not predicted by the causal-inference models and was not consistently expressed at the individual-participant level, it should be interpreted cautiously.

Rather than providing strong evidence for asymmetric causal inference, it highlights the sensitivity of duration judgments to stimulus interpretability and task structure.

Conclusions

In this study, we investigated how auditory duration judgments are influenced by conflicting visual duration information under varying levels of sensory uncertainty. Across all experiments, we observed robust, noise-dependent biases toward visual duration, with bias magnitude increasing as auditory reliability decreased, consistent with reliability-weighted multisensory integration principles (Ernst and Banks, 2002; Van Wassenhove et al., 2008; Hartcher-O'Brien et al., 2014). These biases were largely linear across the tested conflict range and showed only modest asymmetries at larger conflicts. By comparing Bayesian causal-inference models with heuristic integration strategies, we found that no single computational mechanism was uniquely supported by the behavioral data: probabilistic cue-switching, causal inference, and forced-fusion models produced highly similar predictions within the experimentally plausible conflict range. This convergence reflects a fundamental limitation of model identifiability for audiovisual duration tasks: when encoding noise is high and behavioral biases remain approximately linear, distinct integration strategies yield near-indistinguishable behaviors. Importantly, these findings do not argue against the plausibility of causal inference in temporal perception, but instead suggest that short-duration audiovisual judgments may often be adequately explained by simpler heuristic strategies that approximate normative behavior without explicitly inferring causal structure (Caruso et al., 2018; Laquittaine and Gardner, 2018; Ni and Ma, 2024). More broadly, our results emphasize that understanding multisensory timing requires not only normative models but also careful consideration of task constraints, and the limits of model identifiability.

Acknowledgments

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Appendix

Model parameters

Task formulations

This section provides the mathematical formulations for the psychometric models used in each experimental task. We describe how duration estimates arise from noisy sensory measurements and how observers make comparison judgments.

Unimodal duration discrimination

In the unimodal tasks (auditory-only or visual-only), participants compared the durations of two sequentially presented stimuli: a standard stimulus with fixed duration (500 ms) and a test stimulus with variable duration determined by an adaptive staircase. We assume that sensory noise is identical for standard and test intervals in logarithmic duration space, consistent with scalar variability. In linear time, this implies that variability scales proportionally with duration.

The participant's task is to decide which interval was longer by comparing internal duration estimates:

$$P(\text{"test longer"}) = P(\hat{s}_t > \hat{s}_s) = P(\hat{s}_t - \hat{s}_s > 0). \quad (9)$$

Sensory encoding in log-space To capture Weber-like scaling of temporal uncertainty (scalar variability), we assume that duration is encoded in logarithmic space and corrupted by additive Gaussian noise. Each duration measurement m is drawn from a Gaussian distribution centered on the log-transformed true stimulus duration:

$$m \sim \mathcal{N}(\log s, \sigma^2). \quad (10)$$

Under this encoding, measurements in physical time are log-normally distributed, such that the standard deviation of perceived duration increases proportionally with the true duration.

Bayesian decoding (Gaussian prior case) We describe the form of the posterior estimate under the assumption that log-duration has a Gaussian prior, as this derivation underlies the precision-weighted combination used in all subsequent models. Let the measurement model be

$$m \sim \mathcal{N}(\log s, \sigma^2), \quad (11)$$

and assume a Gaussian prior over log-duration,

$$\log s \sim \mathcal{N}(\mu_p, \sigma_p^2). \quad (12)$$

Under these assumptions, the posterior over log-duration is Gaussian. Throughout this paper we use *precision* (inverse variance) to simplify expressions involving reliability weighting. We write $J = 1/\sigma^2$ for the sensory precision and $J_p = 1/\sigma_p^2$ for the prior precision; more generally, $J_x \equiv 1/\sigma_x^2$ for any subscripted variance. The posterior mean (optimal estimator under squared-error loss in log-duration space) is then

$$\hat{s} = \frac{J}{J + J_p} m + \frac{J_p}{J + J_p} \mu_p, \quad (13)$$

which can be written compactly as

$$\hat{s} = w m + (1 - w) \mu_p, \quad w = \frac{J}{J + J_p}, \quad (14)$$

and the variance of the posterior is

$$\sigma_{\hat{s}}^2 = \frac{1}{J + J_p}. \quad (15)$$

Here, all quantities (m , $\hat{s}$, μ_p , σ^2 , σ_p^2) are in log-duration space.

For the standard and test intervals, the posterior means are

$$\hat{s}_s = w m_s + (1 - w) \mu_p, \quad (16)$$

$$\hat{s}_t = w m_t + (1 - w) \mu_p. \quad (17)$$

This formulation assumes that inference is performed in log-duration space, where sensory noise is approximately Gaussian. Performing integration in linear time would not correspond to the MAP solution under this generative model, but rather to a mis-specified inference strategy.

Posterior difference The difference between the two posterior estimates is therefore

$$\hat{s}_t - \hat{s}_s = w (m_t - m_s). \quad (18)$$

Because the same prior is applied to both intervals, the prior mean μ_p cancels out in the comparison. The prior thus acts symmetrically, compressing perceived differences toward zero by the factor $w < 1$, a form of regression toward the mean.

Table 1: Model parameter descriptions

Model	Lapse Rates	Sensory Noise	Prior p_c	Other Params	Total n
Forced Fusion	1	3	–	–	4
Model Averaging	1	3	✓	–	5
Model Selection	1	3	✓	–	5
Probability Matching	1	3	✓	–	5
Probabilistic Cue Switching	1	3	–	$p_{v,l}, p_{v,h}$	6

Note. Sensory noise parameters $\sigma_{a,l}$, $\sigma_{a,h}$, and σ_v represent auditory (low and high noise) and visual standard deviations in log space. Prior parameter p_c is the prior probability of common cause. Other parameters $p_{v,l}$ and $p_{v,h}$ are the switching-probability parameters in low and high auditory noise for the probabilistic cue-switching model.

Decision rule Decisions are based on the comparison of posterior estimates, such that the observer reports “test longer” when $\hat{s}_t > \hat{s}_s$. Under the assumptions above, the posterior difference reduces to

$$\hat{s}_t - \hat{s}_s = w(m_t - m_s),$$

so that the decision is equivalent to comparing the measurements directly, i.e., $m_t > m_s$.

From the modeler’s perspective, the response probability is obtained by marginalizing over the measurement noise:

$$P(\text{“test longer”} \mid s_t, s_s) = P(m_t - m_s > 0). \quad (19)$$

Because m_t and m_s are independent Gaussians with equal variance σ^2 , their difference follows

$$m_t - m_s \sim \mathcal{N}(\log s_t - \log s_s, 2\sigma^2). \quad (20)$$

Therefore,

$$P(\text{“test longer”} \mid s_t, s_s) = \Phi\left(\frac{\log(s_t/s_s)}{\sqrt{2}\sigma}\right), \quad (21)$$

where $\Phi(\cdot)$ denotes the cumulative distribution function of the standard normal distribution.

Psychometric function For fitting purposes, we parameterize the psychometric function with a bias term μ (representing systematic over- or under-estimation in log-space) and a slope parameter σ :

$$P(\text{“test longer”}) = \Phi\left(\frac{\log s_t - \log s_s - \mu}{\sqrt{2}\sigma}\right). \quad (22)$$

Under the assumption of symmetric conditions (same prior, same noise, randomized order), the bias term μ should be zero,

yielding:

$$P(\text{“test longer”}) = \Phi\left(\frac{\log(s_t/s_s)}{\sqrt{2}\sigma}\right). \quad (23)$$

Including a lapse rate λ to account for attentional lapses, the psychometric function used for the unimodal fits becomes:

$$P(\text{“test longer”}) = \frac{\lambda}{2} + (1 - \lambda) \cdot \Phi\left(\frac{\log s_t - \log s_s}{\sqrt{2}\sigma}\right). \quad (24)$$

Cross-modal duration comparison

In the cross-modal task, participants compared the duration of an auditory interval with that of a visual interval and reported which appeared longer. Responses were fitted with a cumulative log-normal psychometric function using maximum-likelihood estimation.

The point of subjective equality (PSE; parameter μ) quantified the modality-specific duration bias, indicating the physical offset required for auditory and visual intervals to be perceived as equal in duration. The slope parameter σ captured discrimination sensitivity.

Posterior means Using the precision notation introduced earlier ($J_a = 1/\sigma_a^2$, $J_v = 1/\sigma_v^2$, $J_{p,a} = 1/\sigma_{p,a}^2$, $J_{p,v} = 1/\sigma_{p,v}^2$), the posterior mean for the auditory estimate in log-duration space is:

$$\mu_{\hat{s},a} = \frac{J_a \cdot \log s_a + J_{p,a} \cdot \mu_{p,a}}{J_a + J_{p,a}}. \quad (25)$$

Similarly for the visual estimate:

$$\mu_{\hat{s},v} = \frac{J_v \cdot \log s_v + J_{p,v} \cdot \mu_{p,v}}{J_v + J_{p,v}}. \quad (26)$$

Posterior variances The variance of the posterior for each modality is the inverse of the sum of precisions:

$$\sigma_{\hat{s},a}^2 = \frac{1}{J_a + J_{p,a}}, \quad \sigma_{\hat{s},v}^2 = \frac{1}{J_v + J_{p,v}}. \quad (27)$$

Decision rule The probability of perceiving the auditory stimulus as longer is:

$$P(\hat{s}_a > \hat{s}_v) = P(\hat{s}_a - \hat{s}_v > 0). \quad (28)$$

The distribution of this difference is:

$$\hat{s}_a - \hat{s}_v \sim \mathcal{N}(\mu_{\hat{s},a} - \mu_{\hat{s},v}, \sigma_{\hat{s},a}^2 + \sigma_{\hat{s},v}^2). \quad (29)$$

Thus, the probability of perceiving auditory as longer is:

$$P(\hat{s}_a > \hat{s}_v) = \Phi\left(\frac{\mu_{\hat{s},a} - \mu_{\hat{s},v}}{\sqrt{\sigma_{\hat{s},a}^2 + \sigma_{\hat{s},v}^2}}\right). \quad (30)$$

Including a lapse rate λ_{cm} for the cross-modal task:

$$P(\text{“auditory longer”}) = \frac{\lambda_{cm}}{2} + (1 - \lambda_{cm}) \cdot \Phi\left(\frac{\mu_{\hat{s},a} - \mu_{\hat{s},v}}{\sqrt{\sigma_{\hat{s},a}^2 + \sigma_{\hat{s},v}^2}}\right). \quad (31)$$

Note that when the priors differ between modalities ($\mu_{p,a} \neq \mu_{p,v}$), a systematic bias emerges even when physical durations are matched ($s_a = s_v$). This is the cross-modal bias that we measured and corrected for in the main bimodal experiment.

Models of audiovisual duration estimation

Forced fusion

The forced-fusion model assumes that participants produce an estimate of auditory duration by integrating visual and auditory measurements in log-duration space, weighted by their relative reliabilities. Under this model, the observer always computes the optimal multisensory estimate:

$$\hat{s}_{\text{fusion}} = \hat{s}_{c=1} = \frac{J_a}{J_a + J_v} m_a + \frac{J_v}{J_a + J_v} m_v, \quad (32)$$

where m_a and m_v denote noisy sensory measurements in log-duration space, and $J_a = 1/\sigma_a^2$, $J_v = 1/\sigma_v^2$ are the corresponding precisions (with $\sigma_a \in \{\sigma_{a,l}, \sigma_{a,h}\}$ depending on condition).

For simplicity, we omit an explicit prior over duration in this expression. In the present 2IFC task, the same duration prior would be applied to both the standard and test intervals, so its

contribution would be largely identical for both and would therefore cancel in the decision variable. By contrast, in the Bayesian causal-inference models below, a bounded duration prior is required to evaluate the marginal likelihoods under the common-cause and separate-causes hypotheses.

The above formula can equivalently be written as a reliability-weighted sum of the measurements:

$$\hat{s}_{\text{fusion}} = w_a \cdot m_a + w_v \cdot m_v, \quad (33)$$

with reliability-based weights:

$$w_a = \frac{J_a}{J_a + J_v}, \quad w_v = \frac{J_v}{J_a + J_v}. \quad (34)$$

Here the forced-fusion model serves as an important benchmark: it predicts that PSE shifts should follow a linear relationship with audiovisual conflict, with slope determined solely by cue reliability. This represents the limiting case where the observer *always* assumes a common cause, regardless of sensory evidence.

Under the common-cause assumption ($C = 1$), the fused estimate corresponds to a Gaussian posterior whose variance is:

$$\sigma_{C=1}^2 = \frac{\sigma_a^2 \sigma_v^2}{\sigma_a^2 + \sigma_v^2} = \frac{1}{J_a + J_v}. \quad (35)$$

In forced-fusion models, this estimate is used on every trial, leading to a linear relationship between audiovisual conflict and bias. In contrast, Bayesian causal-inference models combine this fused estimate with segregated estimates according to the posterior probability of common cause, $p(C = 1 | m_a, m_v)$. Because this posterior probability decreases with increasing conflict, the effective integration strength becomes conflict-dependent, producing nonlinear bias functions.

For forced fusion, the variance of the posterior is always less than the variance of the measurements, $\sigma_{C=1}^2 < \min(\sigma_a^2, \sigma_v^2)$, where $\sigma_a \in \{\sigma_{a,l}, \sigma_{a,h}\}$ demonstrating the benefit of multisensory integration.

Probabilistic cue switching

The probabilistic cue-switching model represents an alternative strategy. On each trial, the observer stochastically selects *one* modality and bases their estimate entirely on that modality’s measurement. This contrasts with averaging-based strategies that combine information from both modalities. The probability of switching between these two cues is a free parameter of the model.

Comparison between switching and integration:

The key behavioral distinction between switching and averaging lies in the distribution of responses across repeated trials at a fixed conflict level. Under switching, at the observer level, the switching model assumes that on each trial the decision is based on either the auditory or the visual measurement. Across trials, this produces a mixture of decision rules. From the modeler's perspective, this mixture results in psychometric functions that can resemble reliability-weighted integration when averaged across trials, even though no cue averaging occurs within a single trial. Averaging models imply a single fused internal estimate on each trial, centered on the reliability-weighted combination of the true stimulus durations. In contrast, switching models imply cue selection on each trial.

Bayesian causal-inference model

Likelihood of a common cause

If the model assumes that both durations arise from a common source, there is a single latent log-duration $y = \log s$, and the measurements (in log-duration space) are:

$$m_a \sim \mathcal{N}(y, \sigma_a^2), \quad (36)$$

and

$$m_v \sim \mathcal{N}(y, \sigma_v^2). \quad (37)$$

Assuming the latent log-duration y has a **flat prior** on $[t_{\min}, t_{\max}]$ (where $t_{\min}$ and $t_{\max}$ denote the bounds of the log-duration range presented in the experiment), this prior is needed to normalize the latent-duration integral and hence to compute the evidence for a common cause:

$$p(y) = \frac{1}{t_{\max} - t_{\min}}, \quad t_{\min} < y < t_{\max} \quad (38)$$

then to find the likelihood of a common cause, we need a single integrand bounded between $t_{\min}$ and $t_{\max}$:

$$p(m_a, m_v | C = 1) = \int_{t_{\min}}^{t_{\max}} p(m_a | y) p(m_v | y) p(y) dy. \quad (39)$$

Let $\Delta_t = t_{\max} - t_{\min}$. Taking the prior out of the integral:

$$p(m_a, m_v | C = 1) = \frac{1}{\Delta_t} \int_{t_{\min}}^{t_{\max}} p(m_a | y) p(m_v | y) dy. \quad (40)$$

Writing the likelihoods as Gaussians:

$$p(m_a, m_v | C = 1) = \frac{1}{\Delta_t} \int_{t_{\min}}^{t_{\max}} \mathcal{N}(m_a; y, \sigma_a^2) \times \mathcal{N}(m_v; y, \sigma_v^2) dy. \quad (41)$$

Because a Gaussian is symmetric in its arguments (mathematically):

$$\mathcal{N}(m_a; y, \sigma_a^2) = \mathcal{N}(y; m_a, \sigma_a^2). \quad (42)$$

It is identical to write the likelihood as:

$$p(m_a, m_v | C = 1) = \frac{1}{\Delta_t} \int_{t_{\min}}^{t_{\max}} \mathcal{N}(y; m_a, \sigma_a^2) \times \mathcal{N}(y; m_v, \sigma_v^2) dy. \quad (43)$$

Product of two Gaussians and final likelihood

The integrand is the product of two Gaussians in the same variable y . By completing the square in y (using precisions J_a and J_v defined earlier), this product factors into a Gaussian in y times a term that is constant with respect to y :

$$\mathcal{N}(y; m_a, \sigma_a^2) \mathcal{N}(y; m_v, \sigma_v^2) = \mathcal{N}(m_a - m_v; 0, \sigma_a^2 + \sigma_v^2) \times \mathcal{N}(y; \mu_c, \sigma_c^2), \quad (44)$$

where the precision-weighted mean and variance of the fused posterior are

$$\mu_c = \frac{J_a m_a + J_v m_v}{J_a + J_v}, \quad \sigma_c^2 = \frac{1}{J_a + J_v}. \quad (45)$$

The first factor, $\mathcal{N}(m_a - m_v; 0, \sigma_a^2 + \sigma_v^2)$, is a constant with respect to y that quantifies the compatibility of the two measurements. Substituting this factorization into the integral and evaluating the Gaussian over the finite prior support gives:

$$p(m_a, m_v | C = 1) = \frac{1}{\Delta_t} \mathcal{N}(m_a - m_v; 0, \sigma_a^2 + \sigma_v^2) \times \left[\Phi\left(\frac{t_{\max} - \mu_c}{\sigma_c}\right) - \Phi\left(\frac{t_{\min} - \mu_c}{\sigma_c}\right) \right], \quad (46)$$

where Φ is the standard normal cumulative distribution function.

Separate causes

Generative model, separate causes Now assuming two separate sources, we estimate two latent durations for each modality (y_a and y_v).

These latent durations y_a and y_v are independent of each other but both are bounded by the same $t_{\min}$ and $t_{\max}$, and they generate measurements from noisy Gaussian distributions:

$$p(m_a|y_a) \sim \mathcal{N}(y_a, \sigma_a^2), \quad (47)$$

$$p(m_v|y_v) \sim \mathcal{N}(y_v, \sigma_v^2). \quad (48)$$

Since they are bounded by the same max/min durations, the uniform prior densities are:

$$p(y_a) = p(y_v) = \frac{1}{\Delta_t} \quad (49)$$

over the range $[t_{\min}, t_{\max}]$.

The likelihood of the two measurements under $C = 2$ is:

$$p(m_a, m_v|C = 2) = \int \int p(m_a|y_a)p(m_v|y_v) \times p(y_a)p(y_v) dy_a dy_v. \quad (50)$$

Factor the double integral Because the integrand separates into a function of y_a times a function of y_v and the domains (boundaries) are independent:

$$p(m_a, m_v|C = 2) = \left[\int_{t_{\min}}^{t_{\max}} p(m_a|y_a)p(y_a) dy_a \right] \times \left[\int_{t_{\min}}^{t_{\max}} p(m_v|y_v)p(y_v) dy_v \right]. \quad (51)$$

Because the integrands are Gaussians in the latent variables, each integral evaluates to a Φ -difference as in the common-cause case. For compactness, define

$$\Delta\Phi_x = \Phi\left(\frac{t_{\max} - m_x}{\sigma_x}\right) - \Phi\left(\frac{t_{\min} - m_x}{\sigma_x}\right), \quad x \in \{a, v\}. \quad (52)$$

The separate-causes likelihood is therefore

$$p(m_a, m_v | C=2) = \frac{\Delta\Phi_a \Delta\Phi_v}{\Delta_t^2}. \quad (53)$$

Posterior probability of common cause

$$p(C = 1|m_a, m_v) = \frac{p(m_a, m_v|C = 1)p_c}{p(m_a, m_v|C = 1)p_c + (1 - p_c)p(m_a, m_v|C = 2)}, \quad (54)$$

where p_c is the prior probability of common cause.

Estimate with model averaging

$$\hat{s} = p(C = 1|m_a, m_v) \cdot \hat{s}_{C=1} + (1 - p(C = 1|m_a, m_v)) \cdot m_a, \quad (55)$$

where $\hat{s}_{C=1}$ is the reliability-weighted fused estimate (in log-duration space):

$$\begin{aligned} \hat{s}_{C=1} &= \frac{\sigma_v^2}{\sigma_a^2 + \sigma_v^2} m_a + \frac{\sigma_a^2}{\sigma_a^2 + \sigma_v^2} m_v \\ &= \frac{J_a}{J_a + J_v} m_a + \frac{J_v}{J_a + J_v} m_v, \end{aligned} \quad (56)$$

where $J_a = 1/\sigma_a^2$ and $J_v = 1/\sigma_v^2$ denote sensory precisions (with $\sigma_a \in \{\sigma_{a,l}, \sigma_{a,h}\}$ depending on condition). In the segregated case ($C = 2$), the observer reports the auditory estimate m_a , consistent with the task instruction to judge auditory duration.

Estimate with probability matching

Probability matching is a stochastic strategy where the observer samples the causal structure from the posterior distribution on each trial:

$$C \sim \text{Bernoulli}(p(C = 1|m_a, m_v)). \quad (57)$$

Then the estimate is determined by the sampled causal structure:

$$\hat{s} = \begin{cases} \hat{s}_{C=1}, & \text{if } C = 1 \text{ (sampled)} \\ \hat{s}_{C=2}, & \text{if } C = 2 \text{ (sampled)}, \end{cases} \quad (58)$$

where $\hat{s}_{C=2}$ is the separate estimate (auditory only in our case):

$$\hat{s}_{C=2} = m_a. \quad (59)$$

Note that while the expected value of probability matching over many trials equals the model averaging estimate (Eq. 8), the two strategies differ in their trial-by-trial behavior. Probability matching produces discrete choices on each trial, whereas model averaging produces a continuous weighted estimate.

Estimate with model selection

Model selection is a deterministic strategy that selects the most probable causal structure and then uses the estimate corresponding to that structure:

$$\hat{s} = \begin{cases} \hat{s}_{C=1}, & \text{if } p(C=1|m_a, m_v) > 0.5 \\ \hat{s}_{C=2}, & \text{otherwise.} \end{cases} \quad (60)$$

This strategy differs from both model averaging and probability matching. Unlike model averaging, which always produces a weighted combination, model selection commits to a single causal structure on each trial. Unlike probability matching, which stochastically samples the causal structure, model selection deterministically chooses the more probable structure. This makes model selection more sensitive to the variance of the posterior: when $p(C=1|m_a, m_v)$ is near 0.5, small changes in sensory evidence can cause discrete switches in behavior.

Model fits to individual subjects' data

Figure 11 shows point of subjective equality (PSE) shifts as a function of audiovisual duration conflict for individual participants, plotted separately for the low and high auditory-noise conditions. Black points with error bars denote empirical PSE estimates derived from psychometric fits. Colored lines indicate model predictions from probabilistic cue-switching, model averaging, model selection, probability matching and forced fusion, using parameters fitted to each participant's bimodal data. This figure illustrates individual differences in bias magnitude and noise dependence, as well as the substantial overlap in predicted PSE–conflict functions across models within the tested conflict range.

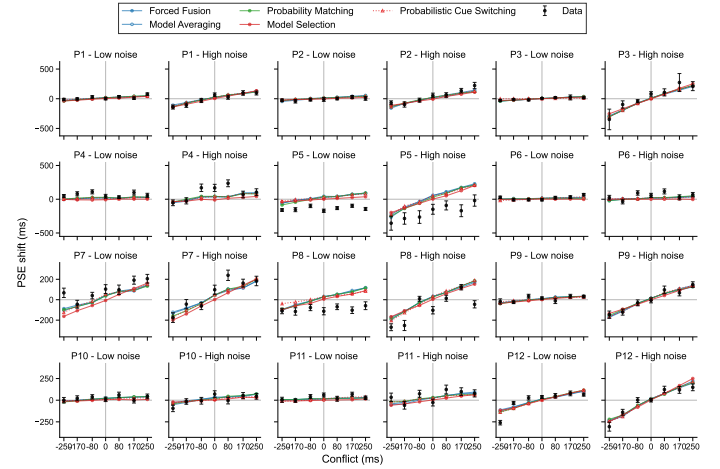


Figure 11: This figure illustrates the relationship between conflict levels and PSE shifts across different auditory-noise conditions for each participant. Here, the σ values were estimated using the bimodal data. Error bars represent the mean 95% confidence interval from bootstrapped data.

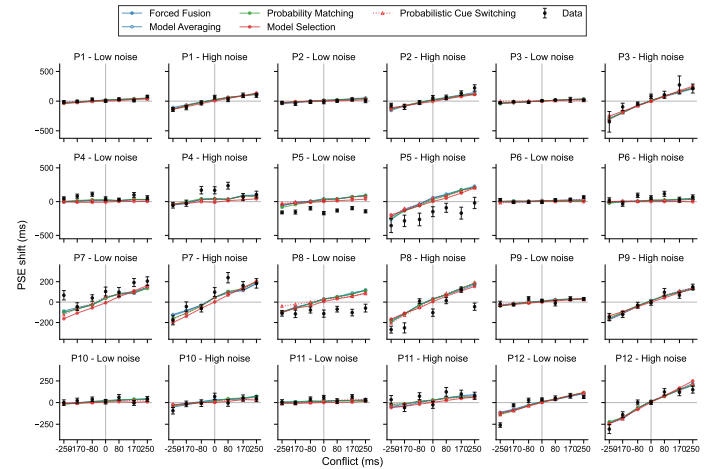


Figure 12: This is exactly the same as Figure 11, except that here the models are fitted using the sensory-noise parameters ($\sigma_{a,l}$, $\sigma_{a,h}$, σ_v) that were estimated based on the psychometric fits to the unimodal data.

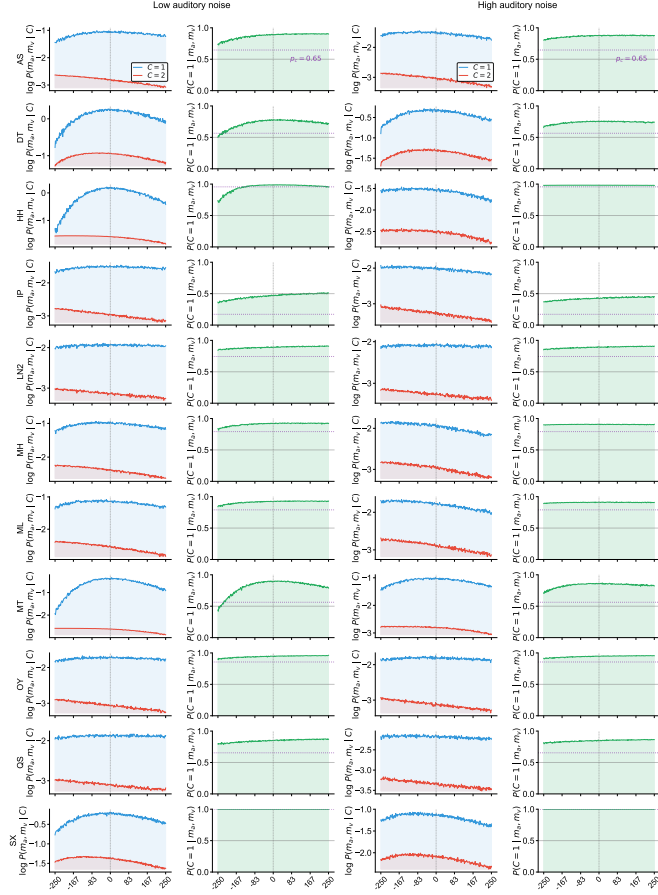


Figure 13: **Log likelihoods and posterior probability of common cause for each participant.** Left two columns: low auditory noise; right two columns: high auditory noise. Columns 1 and 3: the log likelihood of the measurements under a common cause, $\log P(m_a, m_v | C=1)$ (blue), and under separate causes, $\log P(m_a, m_v | C=2)$ (red), are plotted on the same axes. Columns 2 and 4: the posterior probability of common cause, $P(C=1 | m_a, m_v)$, with the dotted line indicating the fitted prior p_c . Each row corresponds to one participant.

Posteriors and likelihoods of common causes

To further illustrate how the causal-inference model interprets audiovisual conflict at the individual-participant level, Figure 13 shows the log-likelihoods of the common-cause and separate-causes hypotheses and the resulting posterior probability of common cause as a function of conflict magnitude under the fitted log-space causal-inference model. Across participants, the likelihood of a common cause generally remained comparable to or higher than that of separate causes, so the posterior probability of common cause stayed relatively high across the tested conflict range, particularly in the high-noise condition. All quantities were computed via Monte Carlo simulation using 2,000 samples

per conflict level and participant-specific sensory-noise estimates (σ_a , σ_v) and common-cause prior (p_c) from the log-normal causal-inference model fit. Across participants, the two log likelihoods decrease at comparable rates with increasing conflict, and the posterior remains close to the prior for nearly all observers, indicating that the fitted parameters leave the causal-inference mechanism unable to distinguish common from separate causes at the conflict levels tested.

Unimodal-derived predictions of bimodal uncertainty

Figure 14 compares empirically observed audiovisual variability (σ_{AV}) with variability predicted under optimal reliability-weighted integration (σ_{opt}), derived from unimodal sensory-noise estimates.

Under the assumption of statistically optimal cue combination with independent Gaussian noise, the predicted audiovisual variance is given by

$$\sigma_{opt}^2 = \frac{\sigma_a^2 \sigma_v^2}{\sigma_a^2 + \sigma_v^2} = \frac{1}{J_a + J_v}, \quad (61)$$

where σ_a and σ_v denote auditory and visual noise, respectively, and $J_a = 1/\sigma_a^2$, $J_v = 1/\sigma_v^2$ are the corresponding precisions.

Observed audiovisual variability (σ_{AV}) was estimated by fitting psychometric functions to bimodal data. Predictions were computed separately for the high- and low-noise auditory conditions using the corresponding unimodal estimates ($\sigma_{a,h}$ and $\sigma_{a,l}$), and compared to the fitted bimodal variability.

Panels show results for each noise conditions. Each point corresponds to an individual participant (color-coded). The dashed unity line ($\sigma_{opt} = \sigma_{AV}$) indicates perfect agreement between predicted and observed variability. Systematic deviations from this line reflect departures from optimal integration based solely on unimodal noise estimates.

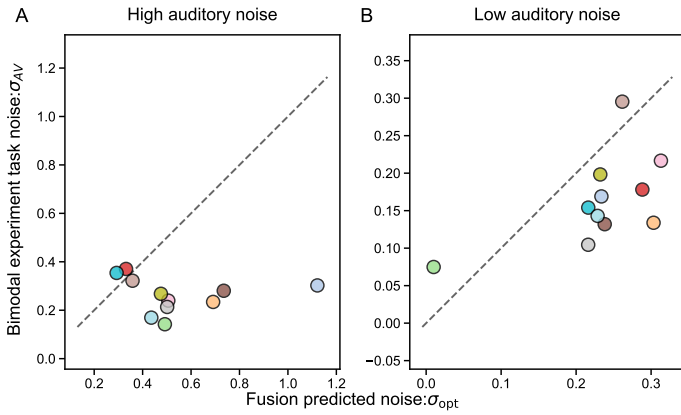


Figure 14: **Comparison of predicted optimal-integration variability with observed audiovisual variability.** Each panel compares empirically observed audiovisual variability (σ_{AV} ; obtained by fitting psychometric functions to bimodal data, y-axis) with variability predicted by optimal reliability-weighted integration (σ_{opt} , x-axis), computed from unimodal psychometric-fit noise estimates ($\sigma_{a,h}$ and $\sigma_{a,\ell}$ for auditory high- and low-noise conditions, and σ_v for vision). Each point is one participant; the dashed line indicates equality ($\sigma_{opt} = \sigma_{AV}$).

Interestingly, observed audiovisual noise was substantially lower than the noise predicted by optimal reliability-weighted integration based on unimodal sensory estimates, particularly in the high auditory-noise condition. In several participants, bimodal uncertainty fell well below the unity line, indicating stronger precision than predicted from unimodal noise alone.

One possible explanation is that the visual stimulus in the bimodal task provided highly precise temporal anchors marking stimulus onset and offset. These visual transients may reduce temporal uncertainty beyond simple duration integration by improving the detection of the interval itself. As a result, auditory duration judgments in the bimodal task may benefit from additional temporal structure (precise cue for onset and offset of the event) that is absent in the unimodal auditory condition, leading to lower overall variability than predicted from unimodal noise estimates alone.

A second possibility is that sensory precision should not be expected to stay stationary across experiment and trials. The bimodal task involved substantially more trials and longer exposure to the stimulus structure, potentially leading to implicit training on temporal discrimination over time. If unimodal noise estimates reflect earlier performance, they may overestimate the effective sensory variability present during the bimodal experiment.

These findings suggest that unimodal-derived sensory noise parameters may not fully generalize to multisensory contexts, particularly when additional temporal cues are available.

Model recovery

Model recovery is a crucial validation step that tests whether our model-comparison procedure can correctly identify the true data-generating model. This ensures that when we select a “best model” for real data, we’re doing so reliably. To perform this analysis, after fitting each candidate model to the empirical data we used the fitted parameter values to generate 200 synthetic datasets per model. Parameters were sampled from group-level distributions estimated from the 11 participants included in the main analyses. For each model and sampled parameter set, when then sampled responses from the model’s predicted psychometric functions. Each synthetic dataset matched the original experimental design (same stimulus conditions, trial counts, and structure). We fitted each model to each of the datasets generated by each model and recorded their AIC values and the winning model for each simulated dataset.

The model-recovery results reveal substantial differences in identifiability across models (Figure 15). Recovery was strongest for the model-selection strategy, which was correctly identified in 88% of simulated datasets, indicating that this model generates relatively distinctive behavioral predictions. Recovery for the probabilistic cue-switching model was more moderate (66%), with appreciable confusion with forced fusion and other averaging-based models. In contrast, forced fusion, model averaging, and probability matching showed poor mutual discriminability, with recovery rates in the range of approximately 32%–41%. This pattern suggests that these models produce similar behavioral signatures and may be difficult to distinguish based on behavioral data alone. Overall, these results indicate that the model-comparison procedure can reliably identify categorical model selection, but provides only limited separation for probabilistic cue switching and for distinctions within the family of averaging-based models.

Parameter recovery

Using the same synthetic datasets, we also conducted a parameter-recovery analysis to confirm that each model’s parameters can be reliably recovered from data generated by that

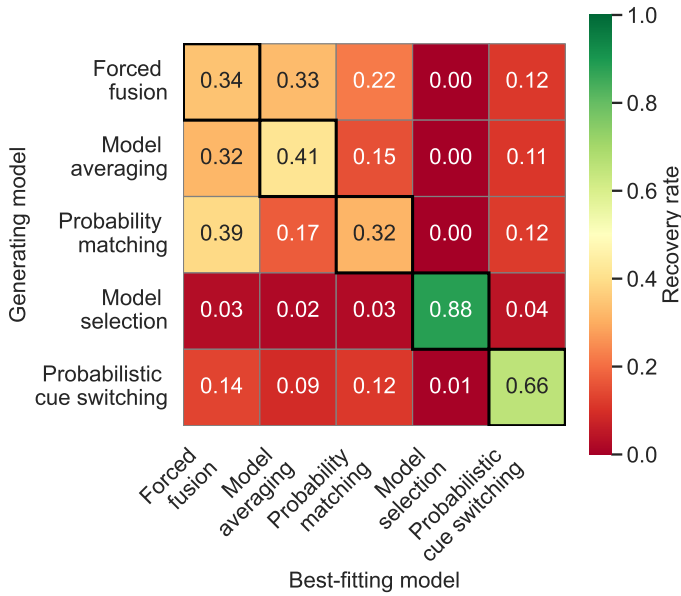


Figure 15: Model recovery. Validation of model identifiability through synthetic data analysis. For each of 5 candidate models (rows), we generated 200 synthetic datasets using parameters sampled from the group-level distribution (mean $\pm$ SD from 11 participants). All 5 models were then fit to each synthetic dataset, and the best-fitting model was determined by AIC. Row-normalized recovery rates show the proportion of times when data generated by one model was identified as coming from each of the five models. Values on the diagonal (outlined in black) indicate successful recovery of the true generating model. The moderate values along the diagonal reveal important differences in model identifiability: selection-based models (model selection, probabilistic cue switching) show good recovery, while averaging-based models (forced fusion, model averaging, probability matching) show poor recovery, indicating that these models make similar predictions and are not discriminable using our behavioral data. This pattern validates model comparison for broad strategy classification (selection vs. averaging) while suggesting caution when making fine-grained distinctions within the averaging family. $n = 200$ iterations per generating model.

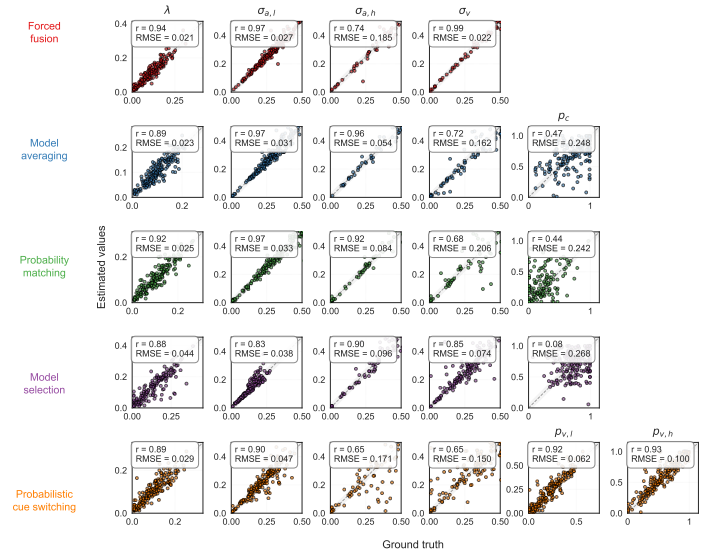


Figure 16: **Parameter-recovery analysis for model validation.** Each row corresponds to a generative model and each column to a model parameter. Synthetic datasets ($N = 200$) were generated from fitted group-level parameter distributions and refit with the same model. Scatter plots show true versus recovered parameter values; dashed diagonals indicate perfect recovery and gray bands show a $\pm 5\%$ tolerance region. Insets report Pearson correlation (r) and RMSE. Sensory-noise and lapse parameters were generally well recovered, whereas causal-prior and switching parameters were less constrained.

model (Figure 16). The parameter-recovery results (Figure 16) demonstrate that model parameters are, in general, well identified. Across models, key sensory-noise parameters ($\sigma_{a,l}$, $\sigma_{a,h}$, σ_v) and the lapse rate (λ) show high correlations between true and recovered values and cluster tightly around the identity line, indicating reliable estimation. In contrast, parameters governing causal inference or strategy selection (e.g., p_c and switching probabilities) exhibit lower correlations and higher variability, reflecting weaker identifiability. This pattern is consistent with the model-recovery results: parameters that strongly shape the psychometric function (e.g., sensory noise) are robustly recovered, whereas parameters that modulate more subtle aspects of behavior are harder to estimate precisely. Importantly, the overall success of parameter recovery confirms that the fitting procedure can reliably capture the main computational components of each model, even in cases where models themselves are difficult to distinguish at a fine-grained level. This dissociation suggests that limitations in model recovery arise primarily from overlapping model predictions rather than from failures of parameter estimation.

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